9 (a) Solve $\frac{4x+3}{2} \geqslant x-3$

$4x + 3 \geq 2(x - 3)$

$4x + 3 \geq 2x - 6$

$2x \geq -9$

$x \geq -4.5$

1 mark
Multiply both sides by positive 2.

1 mark
Collect x terms.

1 mark
Divide by positive 2: keep the inequality direction.

$x \geq -4.5$ **(3)**

(b) Solve $x^2 + 12x + 35 = 0$

$x^2 + 12x + 35 = 0$

$(x + 5)(x + 7) = 0$

$x = -5$ or $x = -7$

2 marks total
Factors multiply to 35 and add to 12.

-5 or -7

(2)

(Total for Question 9 is 5 marks)

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